

PHYSICAL LAYER AND ETHERNET OVERVIEW

LAB 1: PHYSICAL LAYER AND ETHERNET OVERVIEW

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Group 12

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PROCEDURES

This section was structured according to the tasks specified in the project manual. All work was completed using the three lab workstations, two Cisco 3750 Series switches, and one Hewlett-Packard (HP) ProCurve/Aruba switch. Category (CAT) 6 cables were utilized to establish uplink connections, Personal Computer (PC) to switch console connections, and general network connectivity for each switch and PC. To begin, the network architecture was set up by connecting the first workstation to the first Cisco 3750 Series switch, which had been reset beforehand. Static Internet Protocol (IP) addresses were then configured on two Microsoft Windows 11 machines and one Fedora Linux-based machine, and connectivity was confirmed through file sharing, internet access testing, and a printer connection check. Switch management was performed via console connection using PuTTY, during which the Media Access Control (MAC) address table was documented, Virtual Local Area Network (VLAN) assignments were set up and applied, and both start-up and running configurations were recorded. Packet analysis was carried out using Wireshark, during which a Notepad file was shared between PC's so Open System Interconnection (OSI) layers, protocols, and encapsulations could be examined. A Switched Port Analyzer (SPAN) session was then set up to monitor uplink traffic to a designated monitoring PC while visiting the ESPN website. After this, TIA-568A/B straight-through and crossover cables were built and verified using a modular tester. The second Cisco 3750 Series switch along with the HP ProCurve/Aruba switch were reset and configured according to the specifications outlined in the project manual, with proper cabling and PC configurations applied throughout. Bidirectional communications was then assessed under varying Network Interface Card (NIC) speed and duplex settings, using 1 and 100 Megabyte (MB) file transfers to compare actual transfer times against theoretical calculations. The MAC address of one of the

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workstations was subsequently reconfigured. Finally, a large 300 MB file was transferred between workstations, and the resulting Ethernet frames were analyzed in Wireshark to track the payload size along with the number of errored or retransmitted frames. Each task included steps in which could be readily reproduced, as presented in a numbered list format with step-by-step instructions detailing how to complete each one. The following formatting conventions were used throughout: buttons are indicated in **bold**, options are shown in *italics*, text to be entered into the computer in Courier New, and menu navigation is denoted using italicized words separated by pipe symbols (Start | Settings | System | About).

Cisco 3750 Switch Reset and Connection

This section outlines the steps involved in establishing the initial connection between the local workstation and the Cisco 3750A switch, as well as resetting the Cisco 3750A switch itself. Since the Wide Area Network (WAN) connection had already been established on the Rack Switch beforehand, only a cable connection was required to complete the setup. The switch was then reset in order to provide network connectivity to the workstation and enable the initial configuration process. These steps were carried out physically on both the Cisco 3750A switch and the workstation. The appropriate cabling architecture, including the cross connections and patch panels on the rack, is described and can be located in Appendix B.

1. Held Mode on switch for 15 seconds
2. Right-clicked **PuTTY** on PC1
3. Selected *Special command Break*
4. Entered `flash_init`
5. Entered `delete flash:vlan.dat`
6. Entered `delete flash:config.text`

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7. Entered `copy running-config start-up config`

Serial Connection on PuTTY

This section describes the steps used to create a connection to the switch using PuTTY. The process began with starting a console session and reaching the active configuration prompt on the switch. There the management IP address for VLAN 1 the subnet mask and the default gateway were configured. After, GigabitEthernet3/0/3 was enabled so the interface could carry traffic. All the steps were performed directly on the switch using console access via PuTTY.

1. Opened PuTTY
2. Clicked **Serial | Connect | No | No | No | No**
3. Typed `Ctrl-Z`
4. Entered a random key to enable switch serial
5. Typed `en`
6. Typed `conf t`
7. Typed `interface vlan 1`
8. Typed `ip address 44.34.12.12 255.255.255.0`
9. Typed `exit`
10. Typed `ip default-gateway 44.34.12.1`
11. Typed `interface GigabitEthernet3/0/3`
12. Typed `no shutdown`

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Static IP Configuration on Fedora Linux

This section outlines the steps taken to configure the static IP address on the linux-based machine. PC2 was designated as the linux machine and standard IP configurations were followed using the appropriate LAN address. The steps outlined below are completed on PC2

1. Clicked **System Settings | WiFi & Internet | WiFi & Networking | New Connection | Ethernet**
2. Entered 44.34.12.15 in IP Address field
3. Entered 44.2.1.44, 44.2.1.45 in DNS Address field
4. Entered 44.34.12.1 in Gateway field
5. Clicked **OK**

Microsoft Windows 11 Static IP Configurations

This section outlines the steps taken to configure static IP addresses on the two Microsoft Windows 11 machines. According to the project architecture, PC1 and PC3 were designated as the Microsoft Windows 11 machines. Standard IP configurations were then followed, with the same steps being repeated on PC3 as PC1 using the appropriate Local Area Network (LAN) address. The steps outlined were done on PC1

1. Clicked **Start | Settings | Network & internet | Edit | IPv4**
2. Entered 44.34.10.11 in IP Address field
3. Entered 255.255.255.0 in Subnet Mask field
4. Entered 44.34.10.1 in Default Gateway field
5. Entered 44.2.1.44 in Preferred DNS Address field
6. Entered 44.2.1.45 in Alternate DNS Address field

Connectivity Verification

The following describes how all three workstations were configured correctly for network connection after IP address setup. Multiple ways were used to confirm the network connection of each machine, including workstation-to-workstation communication, Internet availability, DNS resolution and access to the lab's printer. Additionally, file sharing between machines was configured and tested to confirm the machines could share files with one another.

1. Typed `ping 44.34.12.15`
2. Typed `ping 44.34.12.17`
3. Typed `ping 8.8.8.8`
4. Typed `nslookup google.com`
5. Typed `ping 44.3.1.206`
6. Right-clicked **Folder**
7. Clicked **Properties | Sharing**
8. Selected *Everyone*
9. Clicked **Share | File Explorer** on PC2
10. Entered `//44.34.12.12/`

Switch Console Configuration and Documentation

This section describes the steps taken to access and document the configuration of the Cisco 3750A switch using a console connection. The switch was managed through PuTTY. The MAC address table, VLAN assignments and both the running configuration and startup configuration were recorded. The VLAN 1 interface on the switch was also set up with a management IP address. The switch was now able to be accessed over the network. All steps were performed on the switch using console access, from PC1.

1. Clicked **PuTTY**
2. Typed `enable`
3. Typed `show mac address-table`
4. Typed `show running-config`
5. Typed `show startup-config`
6. Typed `show vlan brief`
7. Typed `configure terminal`
8. Typed `interface vlan 1`
9. Typed `ip address 44.34.12.12 255.255.255.0`
10. Typed `no shutdown`
11. Typed `exit`
12. Typed `ip default-gateway 44.34.XX.1`
13. Typed `exit`
14. Typed `copy run start`
15. Clicked **Enter**

Packet Analysis - File Sharing

The following steps detail the steps to capture packets with Wireshark. The text file created previously was moved to the FTP server. The move was examined and recorded in Wireshark. The move started from PC1 and the file was opened and downloaded on PC2. Because the file was already on the FTP server, before the capture all the steps following happened on PC2.

1. Clicked **Notepad**
2. Entered `Text`
3. Repeated step 2, 3000 times
4. Clicked **Save**
5. Entered `sample_text`
6. Clicked **Wireshark**
7. Selected *Network Interface*
8. Clicked **Start Capture | File Explorer**
9. Entered `44.34.12.12`
10. Clicked **sample_text | Download | Stop Capture**
11. Entered `tcp.port == 445` in filter field
12. Right-clicked **Packet**
13. Clicked **Follow | TCP Stream | File | Save As**

Span Configuration

This section describes the steps used to set up a SPAN session on the Cisco 3750A switch. The goal is to monitor traffic on the uplink port. PC1 was set up to receive a mirrored copy of all traffic going to and from the uplink while PC3 generated the traffic by visiting espn.com . After analyzing the captured data, the SPAN session was removed. The steps in the process were carried out on the switch through the console connection. The same steps were also carried out on both the monitoring PC and the non-monitoring PC.

1. Opened PuTTY
2. Typed `en`
3. Typed `conf t`
4. Typed `GigabitEthernet3/0/1 both`
5. Typed `monitor session 1 destination interface`
`GigabitEthernet3/0/3`
6. Typed `show monitor session 1`
7. Opened Wireshark on PC1
8. Clicked **Start Capture**
9. Opened web browser on PC3
10. Entered `http://espn.com`
11. Clicked **Stop Capture**
12. Entered `no monitor session 1`

Cable Construction and Verification

This section outlines the steps used to build and check a straight-through cable and a crossover cable with CAT 6 stock and the TIA-568A/B wiring standards. Each straight-through cable and each crossover cable was tested to verify each pin continuity and pin mapping were correct. Documentation of the physical layer characteristics of both the straight-through cable and the crossover cable as well as the data rates and other properties belonging to each cable category was completed.

1. Stripped cable jacket on the first cable
2. Arranged wires per TIA-568B standard
3. Inserted wires into RJ45 connector
4. Crimped connector
5. Repeated steps 4–7 on the second end
6. Inserted cable into modular tester
7. Clicked **Test**
8. Stripped cable jacket on the second cable
9. Arranged wires per TIA-568A standard on one end
10. Arranged wires per TIA-568B standard on the other end
11. Crimped connectors
12. Inserted cable into modular tester
13. Clicked **Test**

Expanded Architecture Setup

This section explains the steps taken to widen the lab network architecture by adding a Cisco 3750 series switch and an HP ProCurve managed switch. Each switch was returned to factory defaults wired according to the layout and given management IP addresses to be accessed over the network. The MAC broadcast domain and contention domain(s) existing in the architecture were then identified and recorded. The steps below were carried out on both switches and on PC2 and PC3.

1. Opened PuTTY
2. Typed `en`
3. Typed `erase startup-config`
4. Typed `reload`
5. Typed `no`
6. Typed `conf t`
7. Typed `interface vlan 1`
8. Typed `ip address 44.34.XX.YY 255.255.255.0`
9. Typed `exit`
10. Typed `ip default-gateway 44.34.XX.1`
11. Opened Command Prompt on PC2
12. Typed `ping 44.34.12.17`
13. Typed `show cdp neighbors`

Line Speed and Duplex Performance Testing

This section describes the steps used to test how well two workstations can communicate in both directions. The tests were done using four line speed and duplex settings. For each setting a 1 MB file and a 100 MB file were sent from one host to the other. The real transfer times were then compared with the expected times based on calculations. The test was done for 100 Mbps duplex, 100 Mbps half-duplex, 10 Mbps full-duplex and 10 Mbps half-duplex configurations. All the steps took place between PC1 and PC3.

1. Right-clicked **Network Adapter**
2. Clicked **Properties | Advanced**
3. Selected *Speed & Duplex*
4. Clicked **OK**
5. Opened Command Prompt
6. Typed `fsutil file createnew file1.txt 1048576`
7. Typed `fsutil file createnew file2.txt 104857600`
8. Copied file1.txt to PC3
9. Recorded transfer time
10. Copied file2.txt to PC3
11. Recorded transfer time
12. Repeated steps 1–12 for remaining speed/duplex settings
13. Repeated steps 1-13 for transferring the files from PC3 to PC1

Line Speed and Duplex Performance Testing Continued

This section explains the steps followed to evaluate file transfer performance between a workstation and an internet-based storage service across the same four line speed and duplex configurations used previously. A 1 MB file and a 100 MB file was transferred to and from the internet storage service at each setting. Both the actual and calculated transfer times were recorded for comparison. The steps described below were performed on PC1.

1. Right-clicked **Network Adapter**
2. Clicked **Properties |Advanced**
3. Selected *Speed & Duplex value*
4. Clicked **OK**
5. Opened web browser
6. Entered Google Drive URL
7. Clicked **Upload**
8. Selected *file1.txt*
9. Recorded transfer time
10. Selected *file2.txt*
11. Recorded transfer time
12. Repeated steps 1–12 for remaining speed/duplex settings
13. Entered results into table

Ethernet Baseline Performance and MAC Address Reconfiguration

This section explains the steps used to create a starting point for Ethernet performance and to see what happens when someone changes a hosts MAC address on purpose. The MAC address table on the switch was examined before and after the reconfiguration. The examination showed how the switch keeps track of which address is connected to which port. A 300 MB file was then sent between two hosts. The number of frames sent, the time taken, and any frames with errors had were analyzed. The steps below occur between PC1 and PC3.

1. Opened PuTTY
2. Typed `show mac address-table`
3. Right-clicked **Network Adapter** on PC1
4. Clicked **Properties | Advanced | Network Address**
5. Entered 001122334455
6. Clicked **OK**
7. Typed `show mac address-table`
8. Opened Command Prompt
9. Typed `fsutil file createnew bigfile.txt 262144000`
10. Copied bigfile.txt to PC3
11. Recorded elapsed time
12. Opened Wireshark
13. Typed `filter`
14. Recorded frame count
15. Recorded error count

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APPENDIX A: Questions and Answers

The following section displayed answered specifics regarding the network configuration of the three switches, with explanations of the several different cable standards utilized to connect different forms of physical media.

Question: What layers of the Open Systems Interconnection (OSI) model are involved in file sharing across the network?

Answer: The process of file sharing implemented all seven layers of the OSI model, where the physical layer transmitted bits over the transmission medium. Frames were transmitted over the data link layer between MAC addresses, and the network layer provided logical addressing through end-to-end communication. The transport layer incorporated the TCP/IP protocol to enable delivery from host-to-host. Even the session, presentation, and application layers were utilized in the formatting and file access of the folder and session.

Question: What protocols were involved during the file sharing process in the LAN environment?

Answer: The Server Message Block 2 (SMB2) protocol was used to network file share between the different machines, where operation was enacted on the final application layer of the OSI model.

Question: Use one packet as an example to explain the how TCP/IP layered architecture and the encapsulation processed.

Answer: The transport layer wrapped the SMB2 data into a segment where the packet had captured the destination port number, which was port 445, and the source port number of 50213. Wireshark displayed a packet with the stated data, as well as the text file name, which proved the SMB2 file data was wrapped in a segment through TCP.

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Question: If someone were only given the captured packets, are they able to reassemble the file transmitted? Use the captured packets to support your answer.

Answer: The file could have been reassembled if every packet holding the transmitted frames were captured and if the packets captured had not been encrypted. Otherwise, the file would not have been able to be reassembled.

Question: What are the physical layer characteristics of the cables you have constructed?

Answer: The constructed cables represented a twisted-pair copper media, which included four wire pairs with modular connectors, as well as a standardized pin assignment. The wire having been a twisted cord, reduced magnetic interference between individual cords and allowed for support when identifying different signals over shorter differences.

Question: Evaluate the physical layer characteristics and resulting data rates of each type of ISO/IEC 11801:2002 standards for “Category” cable.

Answer: When looking at both lower categories, which supported lower bandwidth and data rates and higher categories with improved twist density and insulation reducing any noise. Higher categories enable higher frequencies and therefore higher gigabit-level transmissions over longer distances.

Question: What is the correct term for each of the cables you constructed? Why should the cables not be called “Ethernet cables”?

Answer: The correct terminology was one straight-through and one crossover twisted-pair patch cable. An Ethernet cable refers to a communication standard and therefore should not be used to describe the physical cables in which connected separate physical devices.

Question: What kinds of connections typically use straight-through cables, crossover cables, and rollover cables?

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Answer: The straight-through cable was used to connect different device types, while a crossover cable connected similar device types. The rollover cable was used to serial into a switch for console access.

Question: What IP addresses and network mask were used? Does a default gateway need to be configured?

Answer: The hosts used addresses within the same subnet through the /24 network mask. A default gateway required for hosts on the same local network, but not traditionally for communication purposes.

Question: Assuming these are cables constructed using the TIA-568B standard, what must be done to repair each of these cables?

Answer: To repair a cable of the TIA-568B standard, the cable must have been re-terminated and ordered according to the coloring standard on both ends, ensuring no twisted pair had been reversed or misaligned in the connector.

Question: Assuming these are cables constructed using the ISOC standard, what must be done to repair each of these cables?

Answer: To repair a cable according to the ISOC standard, the cable must have been re-terminated to comply with ISOC wiring standard, ensuring the cable had correct pin placement and twisted pair alignment for reliable data transmission.

Question: Why will a rollover cable not be sufficient for transmission of data over an Ethernet-based network?

Answer: A rollover would not have been sufficient for transmitting data on an Ethernet-based network due to the reverse pin alignment utilized on one end of the cable. The

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cable did not effectively preserve, transmit, or receive pair alignment, which caused the rollover to fall short of providing the signaling needed for data communication.

Question: What Windows Command can create a file with size of exactly 100MB and 1MB?

Answer: The fsutil command on Windows was capable of creating files with specified sizes and names.

Question: How does the calculated time compare to the actual time? Why?

Answer: The calculated transfer time was typically shorter than the actual time due to processing delays and TCP overhead, caused by excess ACK (acknowledgement) packets, which consumed transmission time without holding the file payload. The half-duplex setting was more likely to cause backoff and collision delays.

Question: Examine the MAC address table on the switch and explain the purpose and function of the MAC address table.

Answer: The switch's MAC address table was used to locate physical devices linked to each port on the switch, as the switch used the table for efficiency in not wasting bandwidth on unnecessary ports.

Question: Why would reconfiguring a host be necessary in this manner?

Answer: MAC address-based access control meant only certain device types could gain access to a system, so having changed the value meant someone could bypass security features. Reconfiguring the address also meant new hardware could have been implemented into a system efficiently, without reconfiguring everything pointing to the address.

Question: Is this the actual number of frames transmitted? Explain.

Answer: The actual number of frames transmitted was not equivalent to the calculated value due to retransmission frames and additional traffic the calculated value did not consider.

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APPENDIX B: Configuration of Switches

The following appendix section incorporated three switches and their respective startup/running configuration files. The Cisco 3750A, Cisco 3750B, and HP ProCurve switches were each configured individually. The patch cable management was covered in a table referencing each piece of physical media connected to each switch. A MAC address and VLAN figure were provided for the Cisco 3750A switch to further identify the logical and physical mapping of the architecture. Finally, transfer times were recorded and calculated for the network file share on different duplex settings and file sizes.

Startup-Config output for Cisco 3750A network switch

```
Switch#show startup-config
Using 2900 out of 524288 bytes
!
! Last configuration change at 05:31:01 UTC Tue Apr 5 2011
! NVRAM config last updated at 05:33:11 UTC Tue Apr 5 2011
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Switch
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
switch 3 provision ws-c3750e-48pd
system mtu routing 1500
!
!
!
!
crypto pki trustpoint TP-self-signed-3786916992
  enrollment selfsigned
```

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```
subject-name cn=IOS-Self-Signed-Certificate-3786916992
revocation-check none
rsakeypair TP-self-signed-3786916992
!
!
crypto pki certificate chain TP-self-signed-3786916992
certificate self-signed 01 nvram:IOS-Self-Sig#1.cer
spanning-tree mode pvst
spanning-tree extend system-id
!
!
!
!
!
!
!
!
!
vlan internal allocation policy ascending
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0
no ip address
!
interface GigabitEthernet3/0/1
!
interface GigabitEthernet3/0/2
!
interface GigabitEthernet3/0/3
!
interface GigabitEthernet3/0/4
!
interface GigabitEthernet3/0/5
!
interface GigabitEthernet3/0/6
!
interface GigabitEthernet3/0/7
!
```

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```
interface GigabitEthernet3/0/8
!
interface GigabitEthernet3/0/9
!
interface GigabitEthernet3/0/10
!
interface GigabitEthernet3/0/11
!
interface GigabitEthernet3/0/12
!
interface GigabitEthernet3/0/13
!
interface GigabitEthernet3/0/14
!
interface GigabitEthernet3/0/15
!
interface GigabitEthernet3/0/16
!
interface GigabitEthernet3/0/17
!
interface GigabitEthernet3/0/18
!
interface GigabitEthernet3/0/19
!
interface GigabitEthernet3/0/20
!
interface GigabitEthernet3/0/21
!
interface GigabitEthernet3/0/22
!
interface GigabitEthernet3/0/23
!
interface GigabitEthernet3/0/24
!
interface GigabitEthernet3/0/25
!
interface GigabitEthernet3/0/26
!
interface GigabitEthernet3/0/27
!
interface GigabitEthernet3/0/28
!
interface GigabitEthernet3/0/29
!
interface GigabitEthernet3/0/30
!
interface GigabitEthernet3/0/31
```

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```
!  
interface GigabitEthernet3/0/32  
!  
interface GigabitEthernet3/0/33  
!  
interface GigabitEthernet3/0/34  
!  
interface GigabitEthernet3/0/35  
!  
interface GigabitEthernet3/0/36  
!  
interface GigabitEthernet3/0/37  
!  
interface GigabitEthernet3/0/38  
!  
interface GigabitEthernet3/0/39  
!  
interface GigabitEthernet3/0/40  
!  
interface GigabitEthernet3/0/41  
!  
interface GigabitEthernet3/0/42  
!  
interface GigabitEthernet3/0/43  
!  
interface GigabitEthernet3/0/44  
!  
interface GigabitEthernet3/0/45  
!  
interface GigabitEthernet3/0/46  
!  
interface GigabitEthernet3/0/47  
!  
interface GigabitEthernet3/0/48  
!  
interface GigabitEthernet3/0/49  
!  
interface GigabitEthernet3/0/50  
!  
interface GigabitEthernet3/0/51  
!  
interface GigabitEthernet3/0/52  
!  
interface TenGigabitEthernet3/0/1  
!  
interface TenGigabitEthernet3/0/2  
!
```

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```
interface Vlan1
  ip address 44.34.12.12 255.255.255.0
  !
ip default-gateway 44.34.12.1
ip http server
ip http secure-server
!
!
!
!
!
line con 0
line vty 0 4
  login
line vty 5 15
  login
!
End
```

Startup-Config output for Cisco 3750B network switch

```
Switch#show startup-config
Using 2900 out of 524288 bytes
!
! Last configuration change at 05:25:41 UTC Tue Apr 5 2011
! NVRAM config last updated at 05:39:09 UTC Tue Apr 5 2011
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Switch
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
switch 1 provision ws-c3750e-48pd
system mtu routing 1500
!
!
!
!
crypto pki trustpoint TP-self-signed-3786916992
```

PHYSICAL LAYER AND ETHERNET OVERVIEW

```
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-3786916992
revocation-check none
rsakeypair TP-self-signed-3786916992
!
!
crypto pki certificate chain TP-self-signed-3786916992
  certificate self-signed 01 nvram:IOS-Self-Sig#1.cer
spanning-tree mode pvst
spanning-tree extend system-id
!
!
!
!
!
!
!
!
!
vlan internal allocation policy ascending
!
!
!
!
!
!
!
!
!
!
!
interface FastEthernet0
  no ip address
!
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
```

PHYSICAL LAYER AND ETHERNET OVERVIEW

```
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
!  
interface GigabitEthernet1/0/22  
!  
interface GigabitEthernet1/0/23  
!  
interface GigabitEthernet1/0/24  
!  
interface GigabitEthernet1/0/25  
!  
interface GigabitEthernet1/0/26  
!  
interface GigabitEthernet1/0/27  
!  
interface GigabitEthernet1/0/28  
!  
interface GigabitEthernet1/0/29  
!  
interface GigabitEthernet1/0/30  
!
```

PHYSICAL LAYER AND ETHERNET OVERVIEW

```
interface GigabitEthernet1/0/31
!
interface GigabitEthernet1/0/32
!
interface GigabitEthernet1/0/33
!
interface GigabitEthernet1/0/34
!
interface GigabitEthernet1/0/35
!
interface GigabitEthernet1/0/36
!
interface GigabitEthernet1/0/37
!
interface GigabitEthernet1/0/38
!
interface GigabitEthernet1/0/39
!
interface GigabitEthernet1/0/40
!
interface GigabitEthernet1/0/41
!
interface GigabitEthernet1/0/42
!
interface GigabitEthernet1/0/43
!
interface GigabitEthernet1/0/44
!
interface GigabitEthernet1/0/45
!
interface GigabitEthernet1/0/46
!
interface GigabitEthernet1/0/47
!
interface GigabitEthernet1/0/48
!
interface GigabitEthernet1/0/49
!
interface GigabitEthernet1/0/50
!
interface GigabitEthernet1/0/51
!
interface GigabitEthernet1/0/52
!
interface TenGigabitEthernet1/0/1
!
interface TenGigabitEthernet1/0/2
```

PHYSICAL LAYER AND ETHERNET OVERVIEW

```
!  
interface Vlan1  
  ip address 44.34.12.15 255.255.255.0  
!  
ip default-gateway 44.34.12.1  
ip http server  
ip http secure-server  
!  
!  
!  
!  
!  
line con 0  
line vty 0 4  
  login  
line vty 5 15  
  login  
!  
End
```

Startup-Config output for HP ProCurve

Startup configuration:

```
; J9021A Configuration Editor; Created on release #N.11.78  
  
hostname "ProCurve Switch 2810-24G"  
ip default-gateway 44.34.12.1  
snmp-server community "public" Unrestricted  
vlan 1  
  name "DEFAULT_VLAN"  
  untagged 1-24  
  ip address 44.34.12.16 255.255.255.0  
  exit
```

Figure 2.1: Command to copy Running-config to Startup-config

copy running-config startup-config

PHYSICAL LAYER AND ETHERNET OVERVIEW

Figure 2.2: Table demonstration physical connections established for each PC respectively

Device	Console / Connection	Rack	Patch Panel	Port #	Bench #	X-Connect Port
Cisco 3750A	Uplink	6	(WAN)	N/A	N/A	N/A
PC1	Console	6	2	2	1	Yellow 1
PC1	Connection	6	2	3	1	Green 1
PC2	Console	6	2	4	1	Yellow 2
PC2	Connection	6	2	5	1	Green 2
PC3	Console	6	2	6	1	Yellow 3
PC3	Connection	6	2	7	1	Green 3

Figure 2.3: MAC Address table for Cisco 3750 Switch

Switch#show mac address-table			
Mac Address Table			

Vlan	Mac Address	Type	Ports
----	-----	-----	----
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU

PHYSICAL LAYER AND ETHERNET OVERVIEW

All	0100.0ccd.cdde	STATIC	CPU
All	0180.c200.0000	STATIC	CPU
All	0180.c200.0001	STATIC	CPU
All	0180.c200.0002	STATIC	CPU
All	0180.c200.0003	STATIC	CPU
All	0180.c200.0004	STATIC	CPU
All	0180.c200.0005	STATIC	CPU
All	0180.c200.0006	STATIC	CPU
All	0180.c200.0007	STATIC	CPU
All	0180.c200.0008	STATIC	CPU
All	0180.c200.0009	STATIC	CPU
All	0180.c200.000a	STATIC	CPU
All	0180.c200.000b	STATIC	CPU
All	0180.c200.000c	STATIC	CPU
All	0180.c200.000d	STATIC	CPU
All	0180.c200.000e	STATIC	CPU
All	0180.c200.000f	STATIC	CPU
All	0180.c200.0010	STATIC	CPU
All	ffff.ffff.ffff	STATIC	CPU
1	0011.2233.4455	DYNAMIC	Gi3/0/7
1	0027.0d4c.2203	DYNAMIC	Gi3/0/5
1	40ce.249f.93b7	DYNAMIC	Gi3/0/1
1	8cec.4b84.3be3	DYNAMIC	Gi3/0/5

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1	8cec.4bab.d1f4	DYNAMIC	Gi3/0/3
1	ac7e.8ac3.e392	DYNAMIC	Gi3/0/1
Total Mac Addresses for this criterion: 27			

Figure 2.4: Table demonstrating output for VLAN of configured switch

Switch#show vlan brief			
VLAN Name	Status	Ports	

1 default	active	Gi3/0/1, Gi3/0/3, Gi3/0/5	
		Gi3/0/2, Gi3/0/4, Gi3/0/6	
		Gi3/0/7, Gi3/0/8, Gi3/0/9	
		Gi3/0/10, Gi3/0/11, Gi3/0/12	
		Gi3/0/13, Gi3/0/14, Gi3/0/15	
		Gi3/0/16, Gi3/0/17, Gi3/0/18	
		Gi3/0/19, Gi3/0/20, Gi3/0/21	
		Gi3/0/22, Gi3/0/23, Gi3/0/24	
		Gi3/0/25, Gi3/0/26, Gi3/0/27	
		Gi3/0/28, Gi3/0/29, Gi3/0/30	
		Gi3/0/31, Gi3/0/32, Gi3/0/33	
		Gi3/0/34, Gi3/0/35, Gi3/0/36	
		Gi3/0/37, Gi3/0/38, Gi3/0/39	
		Gi3/0/40, Gi3/0/41, Gi3/0/42	

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	Gi3/0/43, Gi3/0/44, Gi3/0/45
	Gi3/0/46, Gi3/0/47, Gi3/0/48
	Te3/0/1, Te3/0/2
1002 fddi-default	act/unsup
1003 token-ring-default	act/unsup
1004 fddinet-default	act/unsup
1005 trnet-default	act/unsup
copy running-config startup-config	

Figure 2.5: Demonstrating file transfer times between PC1 and PC2

Duplex	Link Speed	File Size	Actual LAN Time	Theoretical Time
Full	100	1	0.13 seconds	0.08 seconds
Half	100	1	0.26 seconds	0.08 seconds
Full	10	1	0.84 seconds	0.8 seconds
Half	10	1	1.49 seconds	0.8 seconds
Full	100	100	8.73 seconds	8 seconds
Half	100	100	12.67 seconds	8 seconds
Full	10	100	84.97 seconds	80.0 seconds
Half	10	100	109.92 seconds	80.0 seconds

Figure 2.6: Demonstrating file transfer times between PC1 and Google Drive

Duplex	Link Speed	File Size	Actual LAN Time	Theoretical Time
Full	100	1	0.95 seconds	0.08 seconds
Half	100	1	1.07 seconds	0.08 seconds
Full	10	1	2.68 seconds	0.8 seconds
Half	10	1	3.36 seconds	0.8 seconds
Full	100	100	21.11 seconds	8 seconds
Half	100	100	30.05 seconds	8 seconds
Full	10	100	128.77 seconds	80.0 seconds
Half	10	100	160.90 seconds	80.0 seconds